EDUSAMBAM ASSESSMENT SERIES

Biology Comprehensive Practice Test 6

PRINTABLE PRACTICE WORKSHEET

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Candidate Name:		**Roll No.:**	
Class / Grade:		**Date:**	
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Subject	Biology	**Topic**	Biology Comprehensive Practice Test 6	**Total Marks**	100
Total Questions	45	**Time Allowed**	1 Hour (60 minutes)	**Version**	A

GENERAL INSTRUCTIONS

1. Read each question carefully before answering.
2. For Section A and Section B, mark one answer only.
3. For Sections C and E, write your answer clearly in the space provided.
4. For Section D, write the correct letter from Column B against each item in Column A.
5. Do not use correction fluid. Keep all answers clear and legible.
6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Column Matching	5 pairs	2	10
Section E — Short Answer	5	4	20
Total	**45**	**—**	**100**

Section A — Multiple Choice — 30 marks

1. Cellular respiration releases energy from glucose mainly as: — Basic · 2 pts

A. Light energy that cells give out
B. Chemical energy in the form of ATP
C. Stored starch in the mitochondria
D. Heat energy that cells cannot use

Answer: ☐A ☐B ☐C ☐D

2. Which is the word equation for aerobic respiration? — Basic · 2 pts

A. Glucose + oxygen → carbon dioxide + water + energy
B. Carbon dioxide + water → glucose + oxygen
C. Glucose → ethanol + carbon dioxide + energy
D. Oxygen + water → glucose + energy

Answer: ☐A ☐B ☐C ☐D

3. Most ATP in aerobic respiration is made in the: — Basic · 2 pts

A. Nucleus
B. Ribosomes
C. Golgi apparatus
D. Mitochondria

Answer: ☐A ☐B ☐C ☐D

4. Anaerobic respiration in human muscle cells produces: — Basic · 2 pts

A. Ethanol and carbon dioxide
B. Glucose and oxygen
C. Lactic acid
D. Starch and water

Answer: ☐A ☐B ☐C ☐D

5. Compared with aerobic respiration, anaerobic respiration releases: **Intermediate · 2 pts**

A. Much less energy per glucose molecule
B. The same amount of energy
C. Much more energy per glucose molecule
D. No energy at all

Answer: ☐A ☐B ☐C ☐D

6. Alveoli are well adapted for gas exchange because they: **Intermediate · 2 pts**

A. Have thick walls to prevent leakage
B. Are far from blood capillaries
C. Have a dry surface
D. Provide a large surface area with thin walls next to capillaries

Answer: ☐A ☐B ☐C ☐D

7. During breathing in (inhalation), the diaphragm and external intercostal muscles: **Intermediate · 2 pts**

A. Relax, so the chest volume decreases
B. Contract, so chest volume increases and the air pressure in the lungs falls
C. Contract, so chest volume decreases
D. Relax, so air is pushed into the lungs

Answer: ☐A ☐B ☐C ☐D

8. Gas exchange in the alveoli moves: **Intermediate · 2 pts**

A. Oxygen out of the blood and carbon dioxide into the blood
B. Oxygen and carbon dioxide both into the blood
C. Oxygen into the blood and carbon dioxide out of the blood
D. Nitrogen into the blood and oxygen out of the blood

Answer: ☐A ☐B ☐C ☐D

9. Which list shows the taxonomic ranks in order from the largest group to the smallest? **Intermediate · 2 pts**

A. Genus, family, order, class
B. Species, genus, family, order
C. Kingdom, class, phylum, order
D. Kingdom, phylum, class, order

Answer: ☐A ☐B ☐C ☐D

10. In the binomial system, the scientific name of an organism consists of: **Advanced · 2 pts**

A. The genus name followed by the species name
B. The family name followed by the order name
C. The kingdom name followed by the genus name
D. A common name followed by the habitat

Answer: ☐A ☐B ☐C ☐D

11. Which feature is shared by all vertebrates? **Advanced · 2 pts**

A. An exoskeleton made of chitin
B. Wings for flight
C. A backbone (vertebral column)
D. Warm blood

Answer: ☐A ☐B ☐C ☐D

12. Fossils provide evidence for evolution because they: **Advanced · 2 pts**

A. Show that species never change
B. Show that organisms in the past differed from living species and record change over time
C. Prove that every species is extinct
D. Are always found in the youngest rock layers

Answer: ☐A ☐B ☐C ☐D

13. Which feature is an adaptation for surviving in a hot, dry desert? **Advanced · 2 pts**

A. Thin, broad leaves with many open stomata
B. Shallow roots that stay close to a river
C. Thick fur that traps heat
D. A fleshy stem that stores water, with leaves reduced to spines

Answer: ☐A ☐B ☐C ☐D

14. Biodiversity loss is most directly reduced by: **Advanced · 2 pts**

A. Protecting habitats and managing species sustainably
B. Removing predators from every habitat
C. Introducing non-native species widely
D. Clearing forests for farmland

Answer: ☐A ☐B ☐C ☐D

15. Which is a main source of new genetic variation in a population? **Advanced · 2 pts**

A. Gaining a skill through practice
B. Eating a different diet
C. Mutation, together with the reshuffling of alleles in sexual reproduction
D. Moving to a new habitat

Answer: ☐A ☐B ☐C ☐D

Section B — True / False — 20 marks

16. Aerobic respiration requires oxygen. **Basic · 2 pts**

A. True
B. False

Answer: ☐A ☐B

17. Breathing and cellular respiration are the same process. **Basic · 2 pts**

A. True
B. False

Answer: ☐A ☐B

18. Anaerobic respiration in yeast produces ethanol and carbon dioxide. **Basic · 2 pts**

A. True
B. False

Answer: ☐A ☐B

19. Gas exchange across the alveoli happens by diffusion. Basic · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

20. Cells respire only when an organism is exercising. Basic · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

21. Humans and chimpanzees share a more recent common ancestor than humans and frogs do. Intermediate · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

22. Organisms with the same common name always belong to the same species. Intermediate · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

23. Individual organisms evolve during their lifetime to meet their needs. Advanced · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

24. Fossils are most commonly found in igneous rock formed from cooled lava. Advanced · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

25. The extinction of species can reduce biodiversity. Advanced · 2 pts
A. True **B.** False
Answer: ☐ A ☐ B

Section C — Fill in the Blank 20 marks

26. The gas given out when yeast respires anaerobically and makes bread dough rise is _____. Intermediate · 2 pts
Answer: _____

27. The molecule that carries usable energy in cells is abbreviated _____. Intermediate · 2 pts
Answer: _____

28. The tiny air sacs in the lungs where gas exchange occurs are called _____. Intermediate · 2 pts
Answer: _____

29. The windpipe is also called the _____. Intermediate · 2 pts
Answer: _____

30. The red pigment in red blood cells that carries oxygen is _____. Intermediate · 2 pts
Answer: _____

31. In a scientific name, the first part, shared by closely related species, is the _____. Advanced · 2 pts
Answer: _____

32. The preserved remains or traces of ancient organisms are called _____. Advanced · 2 pts
Answer: _____

33. The permanent disappearance of a species from Earth is called _____. Advanced · 2 pts
Answer: _____

34. A feature that helps an organism survive and reproduce in its environment is an _____. Advanced · 2 pts
Answer: _____

35. A group of organisms that can breed to produce fertile offspring is a _____. Advanced · 2 pts
Answer: _____

Section D — Column Matching — 10 marks

36–40. Match each term in Column A with the correct description in Column B. **Advanced · 10 pts**

Column A	Answer	Column B
Mitochondrion	____	**A.** Loss of a species from Earth
Alveolus	____	**B.** Two-part scientific name: genus and species
Binomial name	____	**C.** Site of most ATP production
Adaptation	____	**D.** Tiny air sac where gases are exchanged
Extinction	____	**E.** Feature that improves survival and reproduction

Section E — Short Answer — 20 marks

41. Compare aerobic and anaerobic respiration in human muscle cells. **Advanced · 4 pts**

42. Explain how the structure of the alveoli and their blood supply make gas exchange efficient. **Advanced · 4 pts**

43. Explain how the fossil record and the similarity of DNA between species provide evidence for evolution. **Advanced · 4 pts**

44. Why is the binomial naming system useful in biology? **Advanced · 4 pts**

45. Describe two human activities that reduce biodiversity and one action that helps to protect it. **Advanced · 4 pts**

Candidate Signature: _____ **Invigilator Signature:** _____

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Marks Obtained: / 100	Checked By:
Result / Grade:	Date Checked:

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Biology Comprehensive Practice Test 6 — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

1. B
2. A
3. D
4. C
5. A
6. D
7. B
8. C
9. D
10. A
11. C
12. B
13. D
14. A
15. C

Section B — True / False

16. True
17. False
18. True
19. True
20. False
21. True
22. False
23. False
24. False
25. True

Section C — Fill in the Blank (accepted answers)

26. carbon dioxide / co2
27. atp / adenosine triphosphate
28. alveoli / alveolus
29. trachea
30. haemoglobin / hemoglobin
31. genus
32. fossils / fossil
33. extinction
34. adaptation / adaptations
35. species

Section D — Column Matching

36. Mitochondrion — C
37. Alveolus — D
38. Binomial name — B
39. Adaptation — E
40. Extinction — A

Section E — Short Answer (Model Answers)

41. Aerobic respiration uses oxygen to break glucose down completely to carbon dioxide and water, releasing a large amount of energy as ATP. Anaerobic respiration in muscle happens without enough oxygen, breaks glucose down only partly into lactic acid and releases much less energy. Lactic acid can build up and contribute to muscle fatigue.

Marking: 1 mark: aerobic uses oxygen; anaerobic happens without enough oxygen; 1 mark: aerobic releases much more energy (ATP); 1 mark: anaerobic makes lactic acid; aerobic makes carbon dioxide and water; 1 mark: glucose is broken down completely (aerobic) or partly (anaerobic); lactic acid can cause fatigue.

42. The lungs contain millions of alveoli, which gives a very large surface area. Their walls and the capillary walls are only one cell thick, so the diffusion distance is short. A rich capillary network carries blood away and keeps a steep concentration gradient, and breathing keeps bringing fresh air, so oxygen and carbon dioxide diffuse quickly across a moist surface.

Marking: 1 mark: a very large surface area (millions of alveoli); 1 mark: thin walls, so the diffusion distance is short; 1 mark: a rich capillary blood supply keeps a steep concentration gradient; 1 mark: breathing brings fresh air, and the surface is moist.

43. Fossils found in different rock layers show that organisms in the past differed from living species and changed over time. Species with more similar DNA sequences are more closely related and share a more recent common ancestor. These independent lines of evidence agree with each other, which strengthens the case that species have evolved from common ancestors.

Marking: 1 mark: fossils in rock layers show change over time and extinct forms; 1 mark: species share common ancestors; 1 mark: more similar DNA means more closely related; 1 mark: independent lines of evidence agree and support each other.

44. Each species gets one two-part name (genus and species) that scientists use in every country and language, so it avoids the confusion caused by common names, which can differ between places or refer to different organisms. Species that share a genus name are closely related, so the name also hints at relationships.

Marking: 1 mark: the same name is used worldwide, in any language; 1 mark: it avoids confusion caused by common names; 1 mark: it shows relationships (species in the same genus are related); 1 mark: each name has two parts: genus and species.

45. Deforestation and habitat loss remove the places where species live, and pollution, overfishing and hunting reduce populations until some may become extinct. Biodiversity can be protected by creating protected areas or reserves, enforcing laws against poaching, and restoring habitats, so that species keep the habitat and numbers they need.

Marking: 1 mark: one activity, such as deforestation or habitat loss; 1 mark: a second activity, such as pollution, overfishing, hunting or invasive species; 1 mark: a protection action, such as reserves, laws or habitat restoration; 1 mark: explains the effect (loss of habitat, falling numbers, extinction).